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Learning Delay-Coupled Nonlinear Dynamics from functional MRI Data

Master Project Report

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Abstract

Modeling the complex spatiotemporal dynamics of functional Magnetic Resonance Imaging (fMRI) remains a major challenge in computational neuroscience due to regional conduction delays and signal non-stationarity. This project evaluates the capacity of the Koopman operator framework, implemented through the "Koopman" architecture, to identify governing equations from simulated BOLD signals.

The methodology relies on simulations generated via TheVirtualBrain (TVB) using a semi-analytical AdEx mean-field formalism. This environment allows for a controlled study of brain states under fixed biophysical parameters. The Koopa model was implemented and refined through Bayesian hyperparameter optimization using the Mean Squared Scaled Error (MSSE) to account for the low variance of simulated BOLD signals.

Results demonstrate a robust predictive performance, with a Mean Directional Accuracy (MDA) of 69.48% and a Mean Pearson Correlation of 0.51. Internal analysis reveals that while the model successfully implements Koopman-based decomposition, performance remains primarily driven by the encoder-decoder (MLP) capacity. This suggests the Koopman framework serves as a coherent structural foundation, even if its theoretical advantages are partially overshadowed by the dominant role of non-linear mapping for these specific signals.

This work identifies the explicit modeling of inter-regional connectivity and conduction delays as a promising next step to investigate whether their integration, alongside nonlinear dynamics, can provide a more accurate representation of brain activity.

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1 Introduction

1.1 Problematic of multi-scale modeling

The study of the brain as a dynamical system relies on the integration of interactions extending from the synaptic scale to the macroscopic organization of functional networks. At the global scale, the behavior of the system emerges from complex non-linear dynamics between neuronal populations. The major difficulty in modeling lies in dimensionality reduction: it is a matter of capturing the essence of these interactions without being able to individually model each neuronal unit.

To do this, use is made of **mean-field** or **population** models, where it is postulated that the collective activity of a group of neurons can be described by macroscopic state variables. The objective is to define a formal framework allowing to link the physical structure of the network to the observed functional activity.

1.2 Functional Magnetic Resonance Imaging (fMRI)

fMRI is the reference tool for mapping brain activity in a non-invasive manner. Unlike electrophysiological methods, it relies on an indirect measurement: the metabolic and blood response associated with neuronal activity.

1.2.1 The BOLD contrast and neurovascular coupling

The measured signal, called the **BOLD** (*Blood-Oxygen-Level-Dependent*) signal, exploits the magnetic properties of hemoglobin. Deoxygenated hemoglobin locally disrupts the magnetic field, weakening the signal, while oxygenated hemoglobin is magnetically neutral.

During neuronal activation, **neurovascular coupling** triggers a massive influx of oxygenated blood to the active zone. This supply overcompensates for local oxygen consumption, causing a decrease in deoxygenated hemoglobin concentration and, by extension, an increase in MRI signal intensity.

1.2.2 The hemodynamic response (HRF) and its limits

The temporal dynamics of the BOLD signal is described by the **Hemodynamic Response Function (HRF)**. Unlike neuronal electrical activity which is quasi-instantaneous, the HRF acts as a temporal low-pass filter:

- The activation peak is only reached after 4 to 6 seconds.
- The return to baseline requires between 20 and 30 seconds.

This inertia constitutes a major challenge for modeling: the observed signal is a smoothed and delayed version of the real neuronal activity. The goal of this project is therefore to use mathematical tools capable of capturing the dynamic relationships between neuronal populations despite this temporal distortion inherent to the hemodynamic measurement.

1.3 Dynamic characteristics and processing challenges

The analysis of fMRI time series requires addressing complex physical and statistical properties. A fundamental distinction must be made between empirical data, captured from living subjects, and simulated data, generated under controlled mathematical conditions:

- **Non-linearity and Complexity:** In empirical fMRI, the signal is the product of complex neurovascular coupling and non-linear interactions between neuronal populations. These interactions do not follow simple laws of superposition, making the prediction of future states particularly challenging.
- **Noise and latencies:** The signal is altered by a low signal-to-noise ratio and by conduction delays between regions, dictated by the structure and length of white matter bundles.
- **Non-stationarity in Real Data:** Empirical signals are inherently non-stationary. Their statistical properties (mean, variance, correlation) evolve over time due to changes in cognitive states, fluctuations in alertness, or physiological drifts. These transitions reflect the brain's ability to reconfigure its functional organization dynamically.
- **Stationarity in Simulated Data:** In contrast, the simulations used in this project rely on fixed biophysical parameters throughout the generation process. This leads to a form of **quasi-stationarity** where the signal's amplitude and global statistics remain relatively stable. While this simplifies certain aspects of the analysis, it also means the model must focus on capturing the "internal physics" of the system rather than external state transitions.

Faced with these differences, this project explores methods capable of extracting **governing equations** or operator structures. The goal is to determine if models can characterize the underlying dynamics of the signal, whether they originate from the shifting landscapes of real data or the stable, parameter-fixed environment of numerical simulations.

1.4 The spatial unit: from parcellation to population

In this framework, we consider each **Region of Interest (ROI)**, defined by a specific brain **parcellation**, as a mean population unit. While the BOLD signal is physically measured at the **voxel** level, modeling the connectome requires aggregating these voxels into functionally or anatomically coherent parcels.

This approximation is subject to caution: a single ROI aggregates several voxels, and each standard-sized voxel contains tens or even hundreds of thousands of neurons, depending on the specific experimental context (such as the strength of the MRI field, etc.). This approach is more relevant for the **neocortex**, where laminar organization favors local functional coherence within a parcel. Conversely, it is more limited for **subcortical structures**, such as the **thalamus**. The latter is composed of multiple distinct functional nuclei; a broad spatial average across a large ROI would therefore be less representative of the biological reality than a fine-grained parcellation.

1.5 Validation strategy and datasets

The validation of the models will follow a progression in increasing complexity, from controlled simulation to clinical application:

1. **Synthetic data:** The models will first be tested on data generated by biophysical simulators (type *TheVirtualBrain*). These signals present a simplified dynamic and a known "ground truth", allowing to verify the ability of the algorithms to identify the parameters of the system.

2. **Empirical human data:** We will compare recordings in a **resting-state** and under **anesthesia**. Anesthesia offers a system with reduced degrees of freedom, where the dynamics is less complex and more predictable, thus facilitating the identification of transition laws.
3. **Macaque data:** The use of non-human primate data offers an intermediate model, structurally simpler and more robust than human, to test the generalization of our methods.

Study Scope and Data Selection While the initial scope of this project aimed to include empirical validation, the current study focuses exclusively on simulated datasets. This refinement of the project scope was necessitated by the technical complexity of the model calibration phases. This methodological choice ensures a rigorous assessment of the model's performance within a controlled environment where biophysical ground truth is established, providing a necessary foundation before considering future applications to more heterogeneous empirical signals.

2 Theoretical foundations and bibliography

2.1 Computational Neuroscience: Population models and mean-field approximation

2.1.1 From microscopic to mesoscopic scale

Modeling large-scale brain activity, as captured by functional magnetic resonance imaging (fMRI), requires a conceptual and mathematical change of scale. While the fundamental unit of information processing is the individual neuron, an fMRI voxel aggregates the activity of 10^4 to 10^5 neurons. An exhaustive simulation of each unit via detailed biophysical models (such as Hodgkin-Huxley) is numerically prohibitive and makes the identification of global coupling laws complex.

The **population models** approach, or neural mass models, proposes to describe the state of a local assembly by its average statistical variables, such as the *mean firing rate* or the mean membrane potential. This transition to the mesoscopic scale relies on the **mean-field approximation**. Derived from statistical physics, this theory postulates that in a sufficiently large and interconnected network, individual fluctuations of neurons cancel each other out, allowing the collective dynamics to be described by a continuous variable governed by deterministic differential equations.

2.1.2 The Adaptive Exponential Integrate-and-Fire (AdEx) Model

The Adaptive Exponential Integrate-and-Fire (AdEx) model, introduced by [1], serves as a biophysical extension of the classical leaky integrate-and-fire model. It is designed to have a simple mathematical tractability and the phenomenological richness required to reproduce complex neuronal firing patterns, such as adaptation, bursting, and regular spiking.

Mathematical Formulation The model is governed by a system of two coupled nonlinear differential equations describing the evolution of the membrane potential V and an adaptation current w :

$$C \frac{dV}{dt} = -g_L(V - E_L) + g_L \Delta_T \exp\left(\frac{V - V_T}{\Delta_T}\right) - w + I \quad (1)$$

$$\tau_w \frac{dw}{dt} = a(V - E_L) - w \quad (2)$$

When the membrane potential reaches a certain peak value V_{peak} , a spike is emitted, and the variables are reset according to the following rules:

$$\text{if } V(t) > V_{peak} \text{ then } \begin{cases} V \leftarrow V_{reset} \\ w \leftarrow w + b \end{cases} \quad (3)$$

Parameter Significance The dynamics of the AdEx model are determined by several key biophysical parameters:

- **C (Membrane Capacitance):** Determines the time scale of the voltage response to input currents.
- **g_L and E_L (Leak Conductance and Resting Potential):** Represent the passive properties of the membrane, pulling the potential back toward the equilibrium E_L .

- Δ_T and V_T (**Slope Factor and Threshold**): The exponential term models the rapid activation of sodium channels during the initiation of an action potential. V_T is the effective threshold, and Δ_T defines the sharpness of the spike onset.
- w (**Adaptation Current**): A slow outward current that builds up during activity, providing a negative feedback mechanism.
- τ_w (**Adaptation Time Constant**): Governs the decay rate of the adaptation current.
- a (**Sub-threshold Adaptation**): Represents the coupling between the membrane potential and the adaptation current; a higher a increases the resistance to depolarization.
- b (**Spike-triggered Adaptation**): An incremental increase in adaptation after each spike, leading to frequency adaptation (the slowing down of firing over time).

By adjusting these parameters, the AdEx model can emulate a wide variety of neuronal electrophysiological classes. This flexibility, combined with its analytical transparency, makes it the preferred choice for constructing high-fidelity mean-field models that aim to simulate large-scale brain activity while remaining grounded in cellular-level physics.

2.1.3 Semi-analytical Mean-Field Frameworks

While the individual AdEx model provides high biophysical accuracy, simulating large-scale brain dynamics requires a transition to population-level descriptions. Recent advancements have introduced **semi-analytical mean-field** frameworks that derive population transfer functions directly from the biophysical properties of AdEx neurons [2, 3].

The core of this formalism lies in determining the population firing rate ν_{out} as a function of the excitatory and inhibitory input rates (ν_e, ν_i). Unlike purely phenomenological models, this approach relies on a semi-analytical derivation of the neuronal response, capturing the complex relationship between synaptic conductances and the probability of crossing the firing threshold.

First and Second-Order Approximations A key feature of the Zerlaut et al. (2018) [2] formalism is its hierarchical structure based on statistical moments of the membrane potential:

- **First-order approximation:** At this level, the model describes the evolution of the mean firing rate by considering only the mean membrane potential. Interestingly, this first-order approximation is mathematically equivalent to the **Wilson-Cowan model**, providing a direct bridge between biophysical spiking models and classical phenomenological frameworks.
- **Second-order approximation:** The strength of the semi-analytical approach lies in the inclusion of second-order moments, specifically the fluctuations (variance) of the membrane potential σ_V^2 . By accounting for the synaptic noise and the statistical distribution of inputs, the second-order mean-field can reproduce the "high-conductance" states and gain modulations observed in biological brains.

Linking Microscopic Scales to Brain States These models act as a theoretical bridge between *spiking networks* and *whole-brain models*. By linking microscopic parameters—such as adaptation time constants or synaptic decay—to macroscopic dynamics, they allow for the study of global brain states. For instance, transitions between wakefulness and anesthesia can be modeled by adjusting the level of neuromodulatory noise or specific conductance parameters within the mean-field equations.

2.1.4 Large-scale integration and TheVirtualBrain

The objective of this approximation is its integration into a global graph formalism where each node represents a mean-field unit. The global dynamics is then dictated by the structural connectivity (the connectome) which ensures the coupling between these distant populations [4].

The simulation platform *TheVirtualBrain* (TVB) [5] exploits these mean-field models to generate synthetic activity at the whole-brain scale. The transition to the macroscopic level is performed by coupling N population models via a weighted connectivity matrix and by integrating conduction delays related to anatomical distances. This approach allows simulating functional connectivity patterns (similar to fMRI) from rigorous local dynamic laws, thus offering an ideal testing environment for system identification methods.

2.2 Koopman Theory: Global linearization and augmented spaces

2.2.1 Theoretical foundations of the Koopman operator

Paradigm shift: from state space to function space Conventional analysis of non-linear dynamical systems is carried out in a finite-dimensional phase space $\mathcal{M} \subseteq \mathbb{R}^n$. For a discrete-time system, the evolution is governed by a map $f : \mathcal{M} \rightarrow \mathcal{M}$ such that:

$$x_{k+1} = f(x_k) \quad (4)$$

Although this formulation is natural, the complexity of f (non-linearity, instability) often makes global analysis and prediction difficult. In 1931, Bernard Koopman demonstrated that it is possible to represent any finite-dimensional non-linear dynamical system as an infinite-dimensional **linear** system [6]. The fundamental concept consists in shifting the focus from the trajectory of points x toward the evolution of measurement functions, called **observables**, acting on these points [7].

Mathematical definition and linearity Let $\mathcal{F}$ be a Hilbert space of scalar functions (observables) $g : \mathcal{M} \rightarrow \mathbb{C}$. The Koopman operator $\mathcal{K} : \mathcal{F} \rightarrow \mathcal{F}$ is a composition operator defined by:

$$(\mathcal{K}g)(x) = (g \circ f)(x) = g(f(x)) \quad (5)$$

The action of $\mathcal{K}$ on an observable g allows obtaining the value of this measurement at the next time step. The central property of this formalism lies in its **linearity**. Let $g_1, g_2 \in \mathcal{F}$ and $\alpha, \beta \in \mathbb{C}$, we verify that:

$$\mathcal{K}(\alpha g_1 + \beta g_2)(x) = (\alpha g_1 + \beta g_2)(f(x)) = \alpha(\mathcal{K}g_1)(x) + \beta(\mathcal{K}g_2)(x) \quad (6)$$

This linearity is preserved regardless of the complexity of the underlying dynamics f .

Spectral structure and eigenfunctions The analysis of the system relies on the spectral study of $\mathcal{K}$. An eigenfunction $\phi_j \in \mathcal{F}$ associated with an eigenvalue $\lambda_j \in \mathbb{C}$ satisfies the relationship:

$$\mathcal{K}\phi_j = \lambda_j\phi_j \quad (7)$$

This relationship is fundamental for temporal prediction, because the evolution of an eigenfunction along a trajectory is purely linear:

$$\phi_j(x_k) = \lambda_j\phi_j(x_{k-1}) = \lambda_j^k\phi_j(x_0) \quad (8)$$

The eigenvalues λ_j provide information on the stability and periodicity of the system.

Koopman Mode Decomposition (KMD) If we consider the state vector x as a set of observables (the identity $g(x) = x$), we can project this state onto the basis of Koopman eigenfunctions:

$$x_k = \sum_{j=1}^{\infty} \lambda_j^k \phi_j(x_0) v_j \quad (9)$$

where v_j are the **Koopman modes**. This equation allows decomposing the complex activity of fMRI into a superposition of independent linear dynamics, facilitating the interpretation of interactions between regions. This assumes that the state variable x belongs to the span (the space generated) by the operator's eigenfunctions. For non-autonomous or stochastic systems (such as fMRI), this expansion is often an approximation.

2.2.2 Numerical approximation: Extended Dynamic Mode Decomposition (EDMD)

The challenge of finite dimension Although the Koopman operator $\mathcal{K}$ offers a global linear representation, its intrinsically infinite-dimensional nature prohibits a direct analytical solution for the majority of complex systems. In practice, it is necessary to project this operator onto a finite-dimensional subspace $\mathcal{F}_K \subset \mathcal{F}$, defined by a basis of measurement functions called a "dictionary". Global linearization in a finite-dimensional space is only accurate if the function dictionary $\mathcal{F}_K$ defines a subspace that is invariant under the action of the Koopman operator. In practice, with real data, we seek to identify a subspace that is "approximately invariant."

The **Extended Dynamic Mode Decomposition (EDMD)** algorithm, introduced by Williams et al. (2015) [8], generalizes classical DMD by using a dictionary of non-linear observables to more finely capture the non-linearities of the system.

Construction of the dictionary and the augmented space Given a time series of data $x \in \mathbb{R}^n$. We define a dictionary of K basis functions $\boldsymbol{\psi}(x) = [\psi_1(x), \psi_2(x), \dots, \psi_K(x)]^T$. The original state vector is projected into this augmented space:

$$z_k = \boldsymbol{\psi}(x_k) \in \mathbb{C}^K \quad (10)$$

The objective is to identify a finite transition matrix $\mathbf{K} \in \mathbb{R}^{K \times K}$ minimizing the prediction error:

$$\boldsymbol{\psi}(x_{k+1}) \approx \mathbf{K}\boldsymbol{\psi}(x_k) \quad (11)$$

Matrix formulation and least squares resolution To estimate $\mathbf{K}$, two augmented data matrices Ψ_X and Ψ_Y are constructed from M temporal samples:

$$\Psi_X = [\boldsymbol{\psi}(x_1), \dots, \boldsymbol{\psi}(x_M)], \quad \Psi_Y = [\boldsymbol{\psi}(x_2), \dots, \boldsymbol{\psi}(x_{M+1})] \quad (12)$$

The optimal approximation of the operator in the sense of the Frobenius norm is given by [7, 8]:

$$\mathbf{K} = \Psi_Y \Psi_X^\dagger \quad (13)$$

where $\Psi_X^\dagger$ denotes the Moore-Penrose pseudoinverse. This matrix $\mathbf{K}$ then allows extracting the eigenvalues and Koopman modes of the system through simple spectral decomposition.

Limits of the dictionary and utility of Deep Learning The performance of EDMD relies entirely on the ability of the dictionary $\boldsymbol{\psi}$ to span a subspace invariant under the action of $\mathcal{K}$. In the case of fMRI, non-linearities are often unknown and vary between subjects or physiological states. The partially observed and noisy nature of the BOLD signal makes manual discovery of such an invariant subspace extremely difficult. A manually fixed dictionary risks being either incomplete or suffering from the curse of dimensionality. This observation justifies the use of neural architectures to learn the optimal lifting function in an end-to-end manner in order to digitally learn a latent embedding that minimizes the projection error on an invariant subspace, thereby accepting an approximation error inherent to the system. This approach is at the heart of the *Koopa* and *KNF* models.

2.2.3 Deep learning of the operator

The evolution of Koopman theory toward *Deep Learning* responds to the need to learn high-dimensional representations (the *lifting*) without depending on a manual choice of dictionary. The work of Lusch et al. (2018) paved the way by using autoencoders to discover universal

linear embeddings. More recently, the KNF and Koopa models have extended this framework by introducing dynamic correction mechanisms and frequency decomposition to handle systems whose transition laws evolve over time.

A. Universal linear embeddings by deep autoencoders The approach of Lusch et al. [9] constitutes the foundation for learning eigenfunctions through neural networks. The model is based on a modified autoencoder aiming to globally linearize the dynamics.

1. **Autoencoder Architecture:** The model simultaneously learns an encoder $y = \varphi(x)$, a linear transition matrix K such that $y_{k+1} = Ky_k$, and a decoder $x = \varphi^{-1}(y)$. This structure allows identifying intrinsic Koopman coordinates in a *data-driven* manner.
2. **Generalization to continuous spectrum:** A key innovation of this work lies in managing continuous spectrum systems via an **auxiliary network**. This network parametrizes the frequencies of the Koopman operator, allowing for a compact and efficient representation of complex physical dynamics that would otherwise require an infinite-dimensional dictionary.
3. **Interpretability and Parsimony:** By forcing the dynamics to be linear in the latent space, the Lusch et al. model ensures that the discovered representations are physically interpretable in the form of modes, thus laying the groundwork for the prediction and control of highly non-linear systems.

B. Koopman Neural Forecaster (KNF): Inductive Bias and Adaptability The KNF architecture, developed by Wang et al. (2023) [10], is based on the observation that classical deep neural networks (DNNs) struggle to generalize when there are temporal distribution shifts. The model proposes to inject a priori knowledge (inductive biases) via the structure of the Koopman operator.

1. **Encoding and Predefined Measures:** Unlike approaches that learn purely abstract observables, KNF uses a dictionary of canonical measurement functions $\mathcal{G} = [g_1, \dots, g_n]$ (polynomials, sines, exponentials). The originality lies in the use of an encoder ψ that generates not the observables directly, but the coefficients and input parameters of these functions in a latent space. Let X_t be a sequence of observations, the encoder produces a latent matrix V_t such that:

$$G(V_t) = [g_1(v_t^{(1)}), g_2(v_t^{(2)}), \dots, g_n(v_t^{(n)})]^T \quad (14)$$

This structure allows the model to be extremely sample-efficient, as it only needs to approximate the parameters of known physical functions rather than entirely reconstructing the non-linearity.

2. **Triple Operator Architecture:** To model the temporal evolution $z_{t+1} = \mathbf{K}_t z_t$, KNF decomposes $\mathbf{K}_t$ into three components acting in synergy:
 - **The Global Operator ($\mathbf{K}_g$):** A matrix learned statically across all time series in the dataset. It represents the "average" dynamics or fundamental physical laws that do not change (e.g., biological constants of neuronal coupling).
 - **The Local Operator ($\mathbf{K}_{l,t}$):** Learned by a Transformer-type encoder, it captures the specificities of the current time window. It allows the model to adjust to variations in style or dynamic regime specific to an individual or a given moment.

- **The Feedback Adjustment Operator ($\mathbf{K}_{c,t}$):** This is the centerpiece of KNF. This module calculates a correction by comparing the prediction made on the look-back window with the observed reality. If a drift is detected, $\mathbf{K}_{c,t}$ is updated to compensate for the prediction error, acting as a real-time regulator.

C. Koopa: Frequency Disentanglement and Residual Hierarchy The Koopa model, proposed by Liu et al. (2023) [11], introduces an approach based on the separation of time scales. It is inspired by Wold’s decomposition theorem, which suggests that any non-stationary time series can be seen as the superposition of a deterministic component (predictable in the long term) and a purely stochastic component (varying and local).

1. **The Fourier Filter Module:** At the beginning of each block of the architecture, Koopa applies a Fourier Filter. This module transforms the signal into the frequency domain, manipulates amplitudes and phases, and then returns to the time domain to separate the signal into two trajectories:
 - X_{inv} (*Time-invariant*): Low-frequency or cyclic components that represent the stable structure of the system.
 - X_{var} (*Time-variant*): High-frequency residues and sudden drifts that carry the non-stationarity.
2. **Specific Koopman Predictors (KP):** Each trajectory is processed by a dedicated predictor:
 - **Time-invariant KP:** Uses a Koopman operator learned globally by backpropagation. It focuses on the stability of the long-term prediction.
 - **Time-variant KP:** To handle the volatility of X_{var} , this module calculates context-aware operators. Instead of learning a fixed matrix, it locally solves a least-squares problem on the last observed states to find the operator that best describes the immediate transition. This allows Koopa to adapt instantaneously to a change in regime (e.g., the transition from rest to anesthesia).
3. **Hierarchical Learning:** Unlike previous models that relied on heavy reconstruction loss (Autoencoders), Koopa uses a stackable residual structure (inspired by N-BEATS). Each block learns part of the dynamics and passes the residue to the next block. This allows the various Koopman modes to be isolated hierarchically. Optimization is done end-to-end on the forecasting objective, making the model much faster and more memory-efficient.

2.2.4 Application to fMRI dynamics

The application of Koopman theory to fMRI is based on the hypothesis that the complexity of brain interactions can be better understood by changing the representation space.

The ROI as a state variable In our formalism, the activity of a ROI (or a population of neurons) i at time t is defined as a state variable $x_i(t)$. The global state of the system is represented by the vector $\mathbf{x}(t) \in \mathbb{R}^n$, where n is the number of regions of interest. The temporal dynamics of this vector is the result of massive non-linear interactions within the network.

Linearization of couplings through "Lifting" The coupling between neuronal populations is intrinsically non-linear. By projecting voxel activity into a higher dimension (*lifting*) via *Koopa* or *KNF* architectures, we seek to discover a latent space where these complex interactions become linearizable. This change of space allows decomposing BOLD activity into Koopman modes, facilitating the identification of dynamic signatures specific to each region.

Management of non-stationarity and drifts The BOLD signal evolves according to the subject's state (anesthesia vs. rest), posing a challenge of non-stationarity:

- The contribution of *Koopa* [11] is crucial here for capturing changes in dynamic regime during anesthesia induction by isolating the variant components from the baseline dynamics.
- The robustness of *KNF* [10] allows for the management of the distributional shifts inherent in long fMRI recordings, ensuring that the learned transition laws remain stable in the face of measurement noise and physiological drifts.

Architectural Choice: Koopa vs. KNF The selection of the prediction model focused on the **Koopa** architecture, following a theoretical comparative evaluation with the **KNF** (*Koopman Neural Forecaster*) framework. This decision is based on a trade-off between architectural complexity and computational efficiency. Unlike KNF, which relies on heavy MLP, Transformer structures and resource-intensive attention mechanisms, Koopa was designed as an "attention-less" architecture, prioritizing lightweight Multi-Layer Perceptrons (MLPs) and direct mathematical approximations, such as least-squares estimation for the time-variant operator. Given the project's time constraints and the already significant training overhead of the Koopa model, implementing KNF would have exceeded the allocated timeframe. Beyond these technical constraints, focusing on Koopa allows for an assessment of the fundamental viability of Koopman linearization applied to BOLD signals. The objective here is to validate the conceptual effectiveness of the theory and identify its inherent limitations before considering a transition toward more complex and computationally demanding models.

2.3 Graph-based modeling and Neural Differential Equations

2.3.1 The brain as a graph-structured dynamical system

Modeling large-scale brain activity requires taking into account not only the local temporal dynamics of each region but also the complex architecture of the connections linking them. In this context, graph theory emerges as the natural framework for representing the spatial organization of the brain.

Spatial representation and attributes In our study, each voxel or region of interest (ROI) is defined as a **node** ($v \in \mathcal{V}$) within a global graph $\mathcal{G}$. The interactions between these units are represented by **edges** ($\mathcal{E}$), whose weight reflects the strength of structural or functional connectivity. The recorded BOLD signal then constitutes a dynamic attribute attached to each node: at each instant t , the graph is characterized by a feature matrix $H(t)$, where each row corresponds to the signal of a voxel.

Toward continuous dynamics on networks The objective of this approach is to move away from a static view of connectivity to model the propagation of neuronal activity. The brain can then be understood as a structured dynamical system where the evolution of the latent state of a region is constrained by the graph topology [12]. This representation allows for the application of the *Graph Neural ODEs* formalism, treating the brain not as a succession of discrete states, but as a continuous trajectory governed by explicit spatial coupling laws. This perspective is particularly relevant for identifying how global communication pathways are altered during dynamic changes.

2.3.2 Synergy between GNN and ODE: The Graph Neural ODEs framework

Limitations of classical GNNs The article by Liu et al. [12] identifies two major obstacles resolved by the ODE formalism. On one hand, the *forced discretization* of standard GNNs fails to capture the continuity of biological signals. On the other hand, the phenomenon of *over-smoothing* limits the depth of the models. Graph ODEs mitigate these shortcomings by introducing continuous trajectories and physical stability mechanisms.

The integration of differential equations (DE) within graph neural networks (GNN) allows for modeling brain activity no longer as a discrete iteration, but as a continuous flow of latent states.

Mathematical formalism of Graph ODEs In a Graph ODE framework, the hidden state of the nodes $H(t) \in \mathbb{R}^n \times d$ follows a trajectory governed by a drift function f_θ that encodes the structure of the graph $\mathcal{G}$:

$$\frac{dH(t)}{dt} = f_\theta(H(t), t, \mathcal{G}) \quad (15)$$

A classical formulation relies on the graph Laplacian $L = D - A$, modeling propagation as a continuous diffusion process:

$$\frac{dH(t)}{dt} = -\alpha L H(t) W \quad (16)$$

Here, the evolution of a voxel’s state is constrained by the weighted difference with its structural neighbors. Unlike classical GNNs, the function f_θ can be evaluated at any instant t , allowing for arbitrarily fine temporal resolution.

The Unified Architecture: Encoder-Processor-Decoder flow The survey by Liu et al. [12] formalizes the processing according to a modular pipeline:

- **Encoder** (f_{enc}): Defines the initial condition $H(0)$ by projecting the raw BOLD signals X via an initial spatial transformation: $H(0) = \text{GNN}_{enc}(X, \mathcal{A})$.
- **Processor (DE Solver)**: Resolves the trajectory integral over the interval $[0, T]$:

$$H(T) = H(0) + \int_0^T f_\theta(H(t), t, \mathcal{G}) dt \quad (17)$$

The use of adaptive-step numerical solvers (e.g., Runge-Kutta) allows for capturing the complex non-linearities of the signal without the discretization errors inherent in fixed-layer models.

- **Decoder** (f_{dec}): Retrieves the integrated latent state $H(T)$ to predict the future state or reconstruct the signal: $Y = \text{MLP}(H(T))$.

Second-order dynamics and stability To counter over-smoothing, models such as *Graph-CON* use second-order ODEs inspired by damped physical systems:

$$\frac{d^2 H(t)}{dt^2} + \gamma \frac{dH(t)}{dt} = f_\theta(H(t), t, \mathcal{G}) \quad (18)$$

The introduction of inertia (second derivative) and damping (γ) stabilizes training and preserves the spectral richness of the signals, preventing node representations from converging toward a uniform average.

2.3.3 Relevance for the study of non-stationarity

The integration of Graph ODEs offers significant flexibility for the study of biological systems, allowing for an explicit link between brain anatomy and its temporal dynamics.

Capturing spatio-temporal dependencies Unlike Koopman theory, which prioritizes global linearization in an augmented latent space, Graph ODEs keep the **physical topology** of the brain explicit. Since each node of the graph retains its spatial identity (voxel or ROI), the model parameters allow for a direct interpretation of the coupling between regions. As highlighted by Liu et al. (2025) [12], this approach ensures that the propagation of BOLD activity remains constrained by the connectome structure, facilitating the identification of neuronal circuits involved in state transitions.

Management of irregular signals and conduction delays The continuous formalism of ODEs is particularly suited to the constraints of fMRI. On one hand, it naturally handles irregularly sampled data or missing segments. On the other hand, it offers a rigorous framework for modeling **conduction delays** between distant regions by integrating delay variables τ into the drift function:

$$\frac{dH(t)}{dt} = f_\theta(H(t), H(t - \tau), \mathcal{G}) \quad (19)$$

This capability is crucial for distinguishing synchronous interactions from activity propagations across the brain network.

Identification of states and inter-species transitions The transition between wakefulness and anesthesia, or the comparison between the human and macaque brain, is modeled as a change in dynamic regime. This change is mathematically characterized by:

- A modification of the parameters θ of the drift function, reflecting changes in local neuronal excitability.
- A reconfiguration of the graph structure $\mathcal{G}$, highlighting alterations in effective connectivity.

The analysis of trajectories in the parameter space thus provides a precise quantification tool for identifying the dynamic signatures of the loss of consciousness across different scales and species.

Project Scope and Perspectives on GNNs While the integration of Graph Neural Networks (GNNs) and the explicit modeling of transmission delays represent natural extensions for the study of brain dynamics, these aspects were not implemented in the current project. This decision follows a necessary temporal trade-off due to the technical complexity and extended calibration phase required for the previously discussed models. Nevertheless, this report provides a structural analysis of the GNN architectures most suited to this type of data. The objective is to establish a methodological milestone and a reference framework for future work, thereby facilitating performance comparisons between these relational architectures and the models evaluated in this project.

3 Methodology: Simulation and Model Adaptation

3.1 Generation of the synthetic signal using TheVirtualBrain (TVB)

3.1.1 Neural Mass Model Parameterization

Brain dynamics simulations were conducted using TheVirtualBrain (TVB) platform. The selected neural mass model is the mean-field model developed by Zerlaut and al [2]. This approach describes the mesoscopic dynamics of Adaptive Exponential Integrate-and-Fire (AdEx) neuronal networks by integrating conductance-based synapses. This choice allows for a finer representation of cortical activity states, notably the reproduction of the "low-conductance" regimes characteristic of the wakeful state.

The spatial structure of the model is based on a macaque cortex atlas discretized into **68 regions of interest (ROIs)**. The structural connectivity between these regions is defined by a synaptic weight matrix that remains constant across all simulations.

To study the system's sensitivity and generate dynamic variability across brain states, **three key biophysical parameters** were varied within their physiological validity ranges.

- The spike-frequency adaptation current of excitatory neurons (b_e) controls the balance between asynchronous irregular (AI) activity, characteristic of wakefulness, and synchronous Up-Down slow-wave dynamics, characteristic of deep sleep or anesthesia. Increasing b_e progressively destabilizes the self-sustained high-firing state, driving the network into slow-wave oscillations [13].
- The decay time constants of excitatory (τ_e) and inhibitory (τ_i) synaptic conductances modulate network excitability by altering the duration of postsynaptic currents.

Critically, these three parameters do not act independently: for a given combination of τ_e and τ_i , a critical value of b_e exists beyond which the network transitions to slow-wave dynamics, highlighting their interplay in governing brain state transitions [14].

The variation protocol consisted of a one-dimensional sweep: each parameter was modified individually while the others were maintained at their default values. For the adaptation parameter b_e , two distinct random draws ($n_{seed} = 1$ and $n_{seed} = 2$) were performed for each value to evaluate the influence of stochasticity on the stability of the results.

On a numerical level, the simulations initially produce high-resolution raw time series (sampled at 10 points/ms, representing a time step of 0.1 ms). These membrane potential signals are subsequently converted into Blood-Oxygen-Level-Dependent (BOLD) signals via a hemodynamic monitor integrated into TVB (based on the Balloon-Windkessel model). This process simulates neurovascular coupling by filtering high frequencies and introducing physiological latency. The final signal is resampled with a repetition time (TR) of 2 seconds, corresponding to functional MRI standards.

In total, the resulting dataset comprises 42 simulations, representing a raw data volume exceeding 200 GB, thus providing a robust basis for training deep learning models.

3.1.2 Characterization of the obtained time series

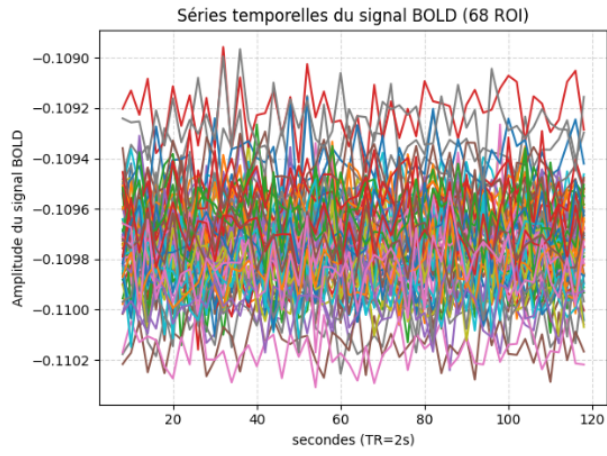
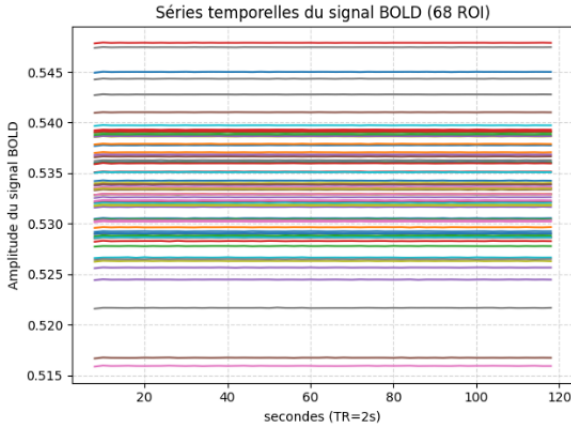
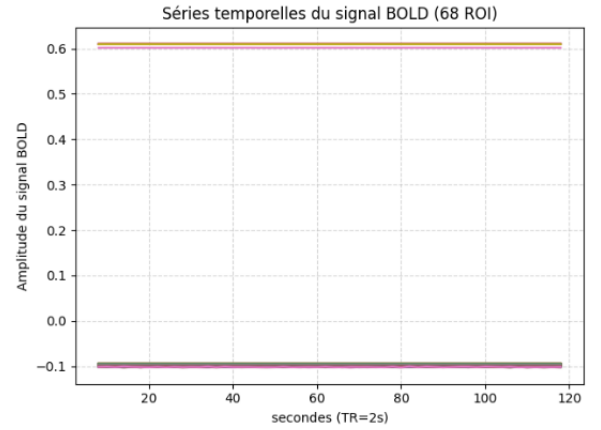


Figure 1: BOLD signal ($b_e=60$, $\tau_{ae}=5$, $\tau_{ai}=9$)



(a) BOLD signal ($b_e=40$, $\tau_{ae}=5$, $\tau_{ai}=5$)

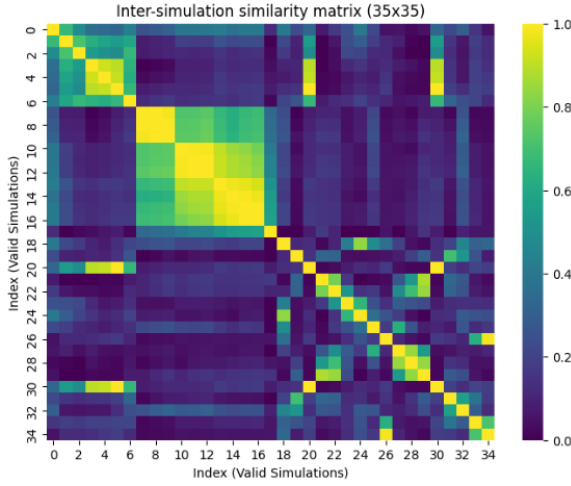


(b) BOLD signal ($b_e=60$, $\tau_{ae}=10$, $\tau_{ai}=5$)

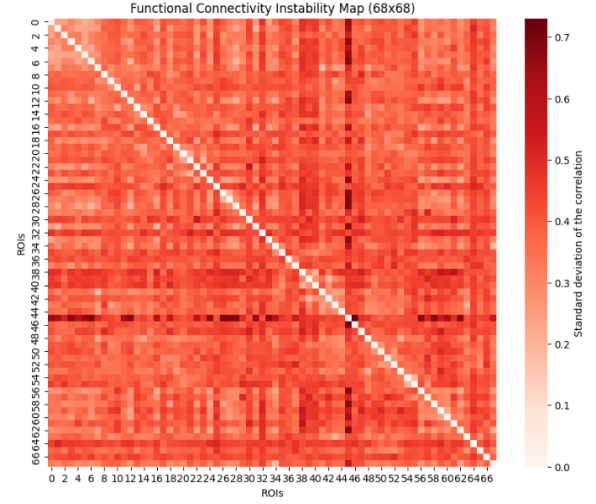
The analysis of the signals generated by the mean-field model of [2] reveals complex dynamics whose properties vary according to the biophysical parameters. This section details the morphological, topological, and dynamical characteristics of the series obtained.

Morphology and heterogeneity of signals The examination of the BOLD signals highlights a strong inter-regional and inter-simulation heterogeneity:

- **Scale and offset heterogeneity:** Major disparities in amplitude are observed at both the inter-regional level (induced by structural connectivity) and the inter-parametric level. Modifications of b_e , τ_e , or τ_i lead to order-of-magnitude changes in average firing frequencies.
- **Impact of standardization:** Despite this diversity, the application of a z -score reveals an underlying structural uniformity. Once normalized, the signals present consistent profiles without saturation zones or total silence, confirming that the model operates in an active dynamic regime across the entire validity range.
- **Intra-simulation stationarity:** With fixed parameters, the series manifest statistical stability (constant mean and variance), characteristic of a system evolving in a quasi-stationary regime.



(a) Inter-simulation similarity matrix Bold signal



(b) Functional Connectivity Instability Map Bold signal

Topology and SC-FC coupling analysis An in-depth study of the relationship between structural connectivity (SC) and functional connectivity (FC) was conducted. The analysis of inter-simulation correlation matrices, including datasets sharing the same macroscopic parameters but distinct initial conditions (random seeds), refutes the hypothesis of a stable correlation "skeleton" imposed by the physical structure. The results show:

- **Coupling instability and local hyper-volatility:** The average similarity of networks (FC) between different simulations drops to 0.232. Mapping the connection variances indicates that the majority of links are unstable. Furthermore, the identification of high-variance bands demonstrates that specific regions, far from acting as stabilizing hubs, see their connections fluctuate massively.
- **Stochastic sensitivity vs. Strong attractors:** The similarity matrix reveals a behavioral dichotomy. In certain parametric ranges, the model converges toward a strong deterministic attractor, resulting in a correlation block close to 1.0 where the network synchronizes identically regardless of the random seed. Conversely, other regimes manifest chaotic sensitivity: a simple modification of the initial noise is sufficient to shift the virtual brain into a radically different functional state.
- **Radical reconfiguration:** These parametric and stochastic variations induce bifurcations that reconfigure the global organization. This observation limits the relevance of a purely static graph convolution approach, as information propagation pathways diverge according to the parametric regime.

Dynamic complexity and metastability The evaluation of the simulated BOLD signals richness shows a dichotomy between local amplitude and network dynamics:

- **Univariate crushing:** Hemodynamic filtering (HRF) significantly smooths fast transitions. The global temporal variance ($\approx 10^{-6}$) and the state transition index ($\approx 10^{-9}$) are numerically low, making "Up/Down" bistability states invisible through simple amplitude observation.
- **Validation through metastability:** Phase dynamics, measured by dynamic Functional Connectivity (dFC), constitute the only indicator of non-trivial complexity. The metastability index shows significant values (mean of 0.23, maximum at 0.38).

These results demonstrate that although the simulated BOLD signal amplitude appears stationary and poor, the system continuously explores new synchronization states. This characteristic is typical of dynamics operating near a critical point, justifying the use of models capable of extracting invariants in an augmented latent space to capture these phase structures.

3.1.3 Identification of Methodological Constraints

The analysis of simulated signal dynamics (Section 3.1.2) identifies several major constraints that dictate the architecture and training of the predictive models.

Signal Heterogeneity and Suppression Observation of the simulated BOLD signals reveals an extremely low variance ($\approx 10^{-6}$) and amplitudes near zero for a large portion of the simulations. This characteristic precludes the use of time series in their raw state:

- **Risk of Trivial Convergence:** A model exposed to signals of such low amplitude tends statistically to predict a zero output to minimize mean squared error, without capturing any real dynamics.
- **Failure of Global Normalization Methods:** Applying global standardization (across the entire dataset) or classic normalization methods risks crushing the biophysical specificities unique to each parameter set. The future model must therefore process each simulation in isolation to preserve its intrinsic dynamics.

Trade-off between Data Volume and Temporal Resolution To address storage and computational power constraints related to the 200 GB of raw data, the decision was made to utilize simulated BOLD time series.

- **Computational Load Reduction:** Transitioning from raw data to BOLD allows for fluid dataset manipulation and accelerated training phases.
- **Sampling Constraints:** However, this choice imposes a significant limitation: each simulation contains only 55 time points. The model must therefore be capable of learning evolution laws from very short sequences, requiring an architecture robust against the risk of overfitting.

Contribution of the Koopman Approach to Metastable Dynamics Although the Koopa model processes each region (ROI) independently, the use of Koopman theory offers a specific advantage regarding the observed metastability:

- **Reconstruction of "Hidden" Dynamics:** The BOLD signal of a single region is a one-dimensional projection of a highly complex system (including neuronal activity, adaptation variables such as w , and the influence of other regions). By projecting this signal into an augmented space (*lifting*), the model seeks to reconstruct a latent space where these complex influences—invisible in BOLD amplitude—follow a predictable linear evolution.
- **Modeling Local Phase Transitions:** Even without visibility of the global network, "lifting" allows for the capture of temporal signatures of phase transitions specific to a region (such as regime shifts induced by metastability). Transforming these non-linear and smoothed patterns into linear operators in the augmented space facilitates the long-term prediction of time series that would otherwise appear random.

Challenges for Spatial Integration via GNNs Utilizing a Graph Neural Network (GNN) model in this context faces the realities of SC-FC coupling:

- **Functional Topology Instability:** The network similarity (0.232) between simulations indicates that the physical structure (SC) does not define a fixed propagation pathway for activity.
- **Constraint on the Graph:** A GNN relying on a static graph or a rigid adjacency matrix would be unable to adapt to the radical network reconfigurations observed according to biophysical parameters. The model must therefore possess sufficient flexibility to integrate this spatial versatility; otherwise, graph convolution might add noise rather than relevant structural information.

3.2 Architecture of the Koopa Model and Specific Adaptations

3.2.1 Core Architecture and Initial Implementation

The prediction model chosen is the Koopa framework (Koopman Forecasting with Purely Attentionless Architecture [11]). As detailed in the theoretical foundations (Section 2.2), this architecture relies on the Koopman operator’s ability to linearize complex dynamics by projecting them into an augmented space, known as the "Koopman space" or lifted space.

The model structure is built around three fundamental characteristics:

- **Auto-encoding and Projection:** A deep auto-encoder is utilized to learn the non-linear mapping from the time series (the 68 regions of interest) to the augmented space, where temporal evolution can be approximated by a linear operator.
- **Hierarchical Time-Variant/Invariant Decomposition:** The signal is decomposed into two distinct branches that respectively process the time-variant and time-invariant components.
- **Residual Learning via Blocks:** The architecture employs a stack of hierarchical learning blocks, where each block processes the residual of the previous block’s prediction, thereby progressively refining the model’s accuracy.

On a technical level, the implementation was carried out based on the official source code provided by the authors of the reference paper. The core architecture of the model was therefore not entirely recoded, which ensured the stability of baseline performance and conformity with results in the literature. The development effort focused on interfacing TheVirtualBrain’s BOLD outputs with the model’s input layers, as well as configuring data pipelines to support specific constraints.

Computational Infrastructure and Implementation The training and optimization of the Koopa architecture, particularly given the high-dimensional nature of the 68 regions of interest (ROIs) and the volume of the TVB simulations, necessitated significant computational power. All implementations and numerical experiments were conducted using the **Data Center for Education (DCE)** at CentraleSupélec [15].

The availability of high-performance CPU and GPU nodes was critical for executing the deep learning pipelines within a feasible timeframe. Specifically, the GPU acceleration allowed for the parallelized processing of the multiple hierarchical learning blocks and the high-frequency sampling required by the AdEx mean-field data. Despite the efficiency of the Koopa framework, the computational cost remained a significant constraint, notably during the hyperparameter optimization (HPO) phase and the processing of extended time series. Without the high-performance resources provided by the DCE, the iterative refinement of the model and the handling of the massive simulation outputs would have been computationally intractable.

3.2.2 Model Adaptations and Modifications

Data Loading Protocol Optimization (Dataloader) The original Koopa architecture utilizes a dataloader designed for continuous time series. The signal is segmented by a sliding window defining a *lookback window* and a *forecast horizon*, the dimensions of which constitute critical hyperparameters.

For our study, this module was modified to ensure data isolation between the different TVB simulations. The partitioning into training, validation, and test sets was restructured to prohibit any overlapping of sliding windows between two distinct parameter sets. This isolation is essential to avoid data leakage and ensures that the model evaluates its generalization capacity on dynamic regimes it has not encountered during training.

Handling Heterogeneity through Reversible Instance Normalization (RevIN) The direct application of the model to raw BOLD time series initially led to a failure in convergence. The heterogeneity of amplitudes and offsets between simulations (described in Section 3.1.2) caused large-scale errors to dominate the loss function, pushing the model toward trivial predictions (zero signal) to minimize error variance.

Two classic strategies were discarded after testing:

- **Training on a single simulation:** Although raw data offer a higher point density, the repetitive nature of their dynamics led to massive overfitting, rendering the model incapable of predicting other parameter sets.
- **Global standardization:** This method excessively standardized the data, causing the model to lose the specificities of each biophysical regime and rendering the de-standardization phase (returning to real units) ineffective.

To address the challenges, a *Reversible Instance Normalization* (RevIN) module [16] was integrated into the model architecture. RevIN operates as an instance-wise normalization–denormalization wrapper applied independently to each input window and each variable.

Given an input time series $x^{(i)} \in \mathbb{R}^{K \times T_x}$, where K is the number of variables (here, the number of ROIs) and T_x the temporal length, RevIN computes instance-specific statistics along the temporal dimension:

$$\mu_k^{(i)} = \frac{1}{T_x} \sum_{t=1}^{T_x} x_{kt}^{(i)}, \quad \sigma_k^{2(i)} = \frac{1}{T_x} \sum_{t=1}^{T_x} (x_{kt}^{(i)} - \mu_k^{(i)})^2 \quad (20)$$

where $\mu_k^{(i)}$ and $\sigma_k^{2(i)}$ denote the mean and variance of variable k for instance i . Importantly, these statistics are computed **independently for each time series instance and each variable**, making the normalization fully instance-wise.

Normalization phase. The input is normalized as:

$$\hat{x}_{kt}^{(i)} = \gamma_k \left(\frac{x_{kt}^{(i)} - \mu_k^{(i)}}{\sqrt{\sigma_k^{2(i)} + \epsilon}} \right) + \beta_k \quad (21)$$

where ϵ is a small constant for numerical stability, and γ_k, β_k are learnable affine parameters.

Denormalization phase. After the model produces a prediction $\tilde{y}^{(i)}$, the original data distribution is restored via:

$$\hat{y}_{kt}^{(i)} = \sqrt{\sigma_k^{2(i)} + \epsilon} \left(\frac{\tilde{y}_{kt}^{(i)} - \beta_k}{\gamma_k} \right) + \mu_k^{(i)} \quad (22)$$

This reversible formulation ensures that the statistical properties removed during normalization are explicitly reintroduced at the output stage.

Comparison with Batch Normalization. Unlike Batch Normalization, which computes global statistics across a mini-batch (shared mean and variance for all samples), RevIN performs normalization at the *instance level*. In Batch Normalization, the estimated statistics can be biased when the batch contains heterogeneous or temporally shifted samples, which is common in non-stationary time series. As a result, Batch Normalization fails to properly align distributions when the data exhibit strong temporal shifts between training and testing phases.

In contrast, RevIN adapts dynamically to each individual input sequence by computing its own statistics. This makes it inherently more robust to distribution shifts, as it removes instance-specific mean and variance discrepancies before prediction. Consequently, RevIN provides a more stable and consistent input space for the model.

Interpretation and advantages. By removing instance-specific non-stationary components prior to prediction, RevIN allows the model to operate in a stabilized feature space with reduced distribution discrepancies across samples. The model can therefore focus on learning intrinsic temporal dynamics rather than compensating for scale and shift variations. Moreover, the reversible design preserves the original data distribution through explicit denormalization, avoiding information loss. This results in improved robustness and significantly enhanced forecasting performance, especially in the presence of strong non-stationarity.

However, this approach relies on the implicit assumption that each input window is approximately stationary. When this assumption is violated, for instance, in the case of very short input windows or strong intra-window distribution shifts, the estimated statistics may become noisy or biased. This can potentially introduce instability in both the normalization and denormalization steps. Nevertheless, in practice, the learnable affine parameters (γ, β) and the model’s focus on relative temporal dynamics mitigate these effects, leading to improved robustness overall.

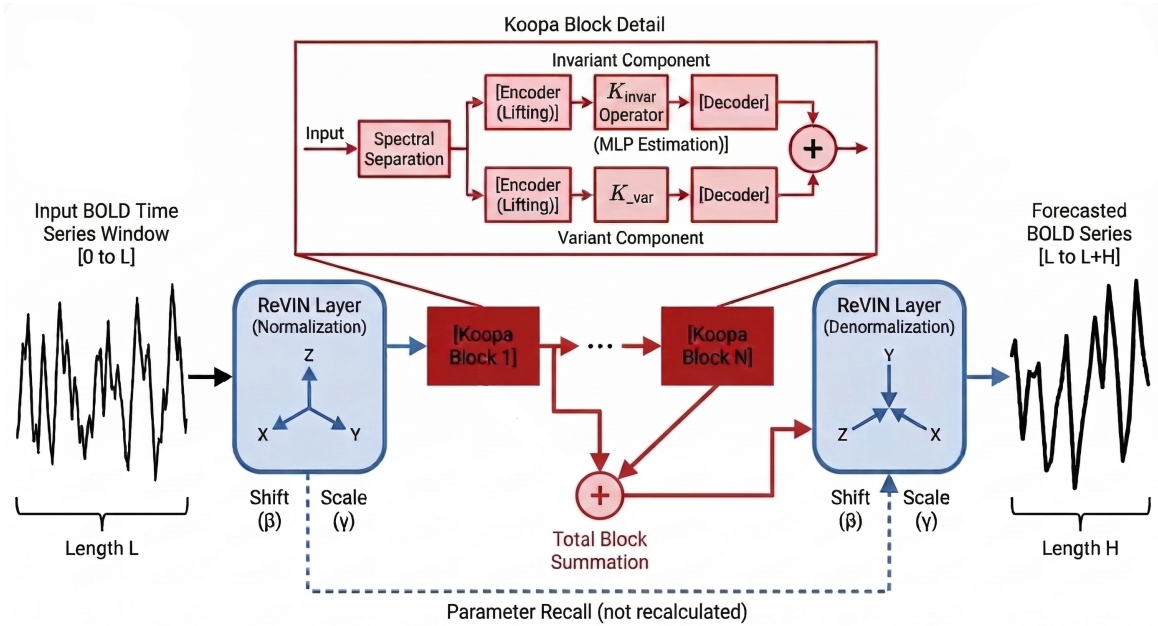


Figure 4: Architecture Koopa + ReVIN

3.2.3 Training Strategy and Metrics

Loss Function Optimization: From MSE to MSSE Initially, the model training relied on a Mean Squared Error (MSE) loss function. However, using the MSE in the normalized space alongside the RevIN module revealed a critical side effect: the learnable parameters of the normalization module tended to converge toward a trivial solution (zero output), artificially minimizing the MSE in the normalized space without capturing any real dynamics. Consequently, it is essential to use the MSE in the original space, at least for the validation loss.

Furthermore, standard MSE proved sensitive to amplitude disparities between simulations, prioritizing accuracy on high-variance signals at the expense of low-amplitude regimes. To mitigate this bias, the Mean Squared Scaled Error (MSSE) was adopted. This metric normalizes the error of each window by a measure of the input signal’s scale, allowing for a fair weighting of errors regardless of absolute amplitude:

$$MSSE = \frac{\frac{1}{H} \sum_{i=1}^H (y_i - \hat{y}_i)^2}{\frac{1}{L-1} \sum_{j=2}^L (x_j - x_{j-1})^2} \quad (23)$$

Where L is the length of the lookback window, H is the prediction horizon, x is the input signal, and $y, \hat{y}$ are the ground truth and predicted values, respectively.

Robustness and Reproducibility To ensure that performance did not result from a favorable stochastic initialization, the HPO and final evaluations were conducted using a fixed multi-seed protocol. This approach guarantees the repeatability of the experiments. However, due to the high computational cost and training duration per simulation, cross-validation was not utilized. The model’s robustness is evaluated here by its ability to generalize to the test set, which consists of simulations and windows not encountered during training.

Final Evaluation Metrics Once the model is trained, its performance on the test set is quantified after de-standardization (returning to original units) according to two axes:

- **Point-to-point reconstruction error:** Using standard metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE).
- **Dynamic and temporal coherence:**
 - *Pearson Correlation Coefficient:* To evaluate the morphological similarity and synchrony between the predicted signal and the ground truth.
 - *Mean Directional Accuracy (MDA):* To measure the model’s ability to correctly predict the direction of variation (increase or decrease) of the signal, a crucial indicator for capturing BOLD signal trends.

3.2.4 Hyperparameter Optimization (HPO) and Sensitivity Analysis

The search for optimal hyperparameters was conducted using **Bayesian optimization** [17], with the **MSSE** metric serving as the scoring function during both training and validation phases.

1. Preliminary Configuration: Setting the Time Frame During the initial exploratory phases, it became evident that the windowing parameters, specifically the prediction horizon (H) and the number of input blocks, saturated the search space, masking the influence of structural parameters (see Figure 6).

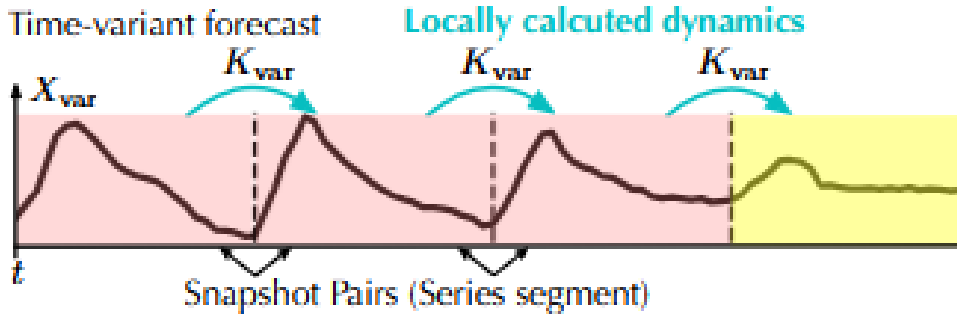


Figure 5: input windows in block and prediction windows (image from [11])

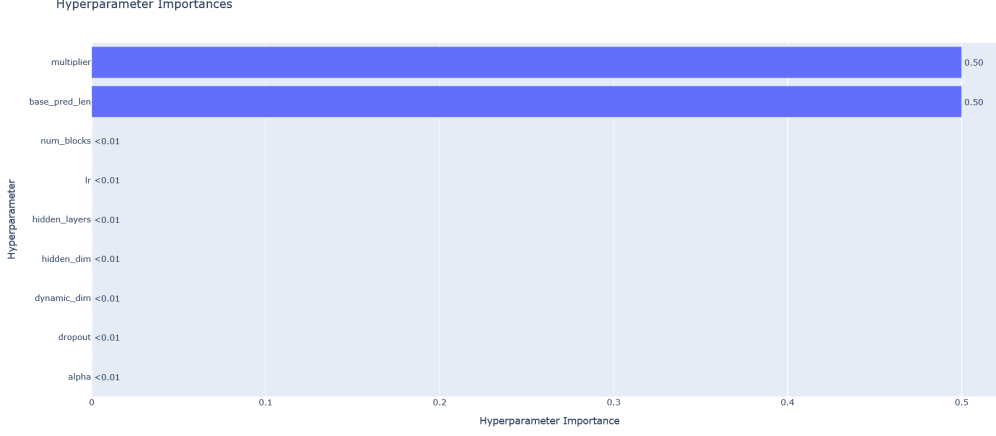


Figure 6: Hyperparameters importance, HPO with windows paramters

To stabilize the optimization process, these temporal variables were fixed to values that ensure a compromise between informational richness and the statistical stability of the dataset:

- **Prediction Horizon (H):** 4 TR (4 points).
- **Lookback Window:** 24 TR (resulting from the use of 6 input prediction blocks).

This configuration maintains a sufficient number of sliding windows per simulation to ensure convergence without inducing excessive sensitivity to local noise due to too small train and validation set.

2. HPO Results and Optimal Configuration The search space and the resulting optimal configuration obtained via the Bayesian optimization process are summarized in Table 1.

Table 1: Hyperparameter search space and optimized values with loss MSSE.

Hyperparameter	Search Space	Optimal Value found
Learning Rate (lr)	$[10^{-5}, 10^{-2}]$ (Log)	2.77×10^{-4}
Dropout	$[0.1, 0.5]$ (Step 0.05)	0.30
Dynamic Dimension	$\{16, 32, 64, 128, 256\}$	128
Hidden Dimension	$\{32, 64, 128, 256, 512\}$	512
Hidden Layers	$\{1, 2, 3, 4\}$	2
Number of Blocks	$\{1, 2, 3, 4\}$	3
Alpha (α)	$[0.1, 0.5]$ (Step 0.05)	0.15

3. Analysis of Hyperparameter Importance The importance graph (see Figure 7) summarizes the contribution of each hyperparameter to the minimization of the MSSE loss. The analysis reveals a clear hierarchy where raw representation capacity prevails over mechanisms specific to Koopman theory.

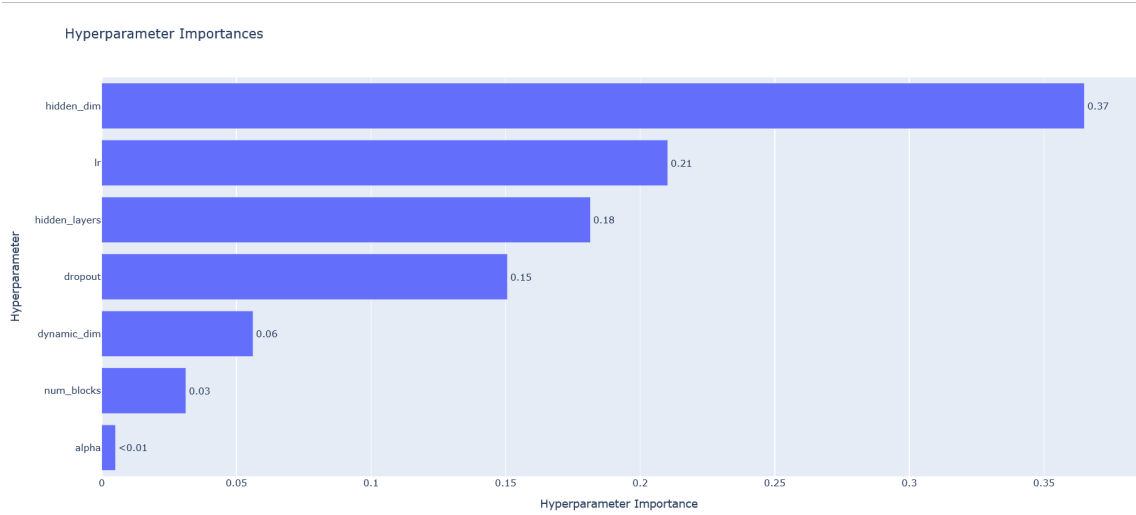


Figure 7: Hyperparameters importance, HPO with windows parameters fixed, loss MSSE

- i) **Domination of Parametric Capacity (MLP):** The parameters `hidden_dim` (0.37) and `hidden_layers` (0.18) account for **55% of the global importance**. The model's performance is primarily governed by the dimensionality of the dense networks (encoders, decoders, and operators). This result suggests that the model favors a raw non-linear approximation via a Multilayer Perceptron (MLP) rather than a fine exploitation of the dynamic structure imposed by the architecture.
- ii) **Marginalization of Koopman Components:** Contrary to theoretical expectations, parameters dedicated to the Koopman operator show marginal influence:
 - `dynamic_dim` (0.06): The dimension of the latent ("lifted") space is not a limiting factor. This corroborates the idea that the invariant operator processes a negligible fraction of the simulated BOLD signal.
 - `num_blocks` (0.03): The hierarchical stacking of residual dynamics blocks does not yield significant improvement, indicating that dynamic complexity is not captured iteratively by the Koopa structure.
- iii) **Ineffectiveness of Spectral Separation:** The `alpha` parameter (< 0.01), intended to drive the separation between variant and invariant components via Fourier transform, shows no influence. The model is entirely insensitive to the spectral cutoff threshold, provided that it is not zero, questioning the relevance of this decomposition for simulated BOLD signals where dynamics are already smoothed by the hemodynamic response.
- iv) **Sensitivity to Optimization and Overfitting:**
 - `lr` (0.21): The learning rate is the second most critical parameter, revealing a complex loss landscape with a high risk of stagnation or divergence.
 - `dropout` (0.15): The importance of dropout confirms the model's tendency toward overfitting. By relying on dense networks (high `hidden_dim`), the system becomes vulnerable to memorizing noise within the simulated time series.

4. Synthesis of the HPO The HPO analysis reveals a hierarchy of influence where the encoder-decoder capacity (MLP) acts as the primary driver of the model’s predictive accuracy. This dominance suggests that high-capacity non-linear mapping is essential to project complex simulated BOLD signals into a functional latent space. However, this does not marginalize the role of the Koopman framework. Rather, it indicates that deep learning provides the necessary flexibility for the underlying theory to operate on such heterogeneous data. While parameters like the lifting dimension appear secondary in the optimization process, they provide the structural backbone allowing the model to bridge deep learning flexibility with operator theory.

4 Performance Analysis and Discussion

4.1 Quantitative Evaluation of Results

4.1.1 Reconstruction and Prediction Accuracy

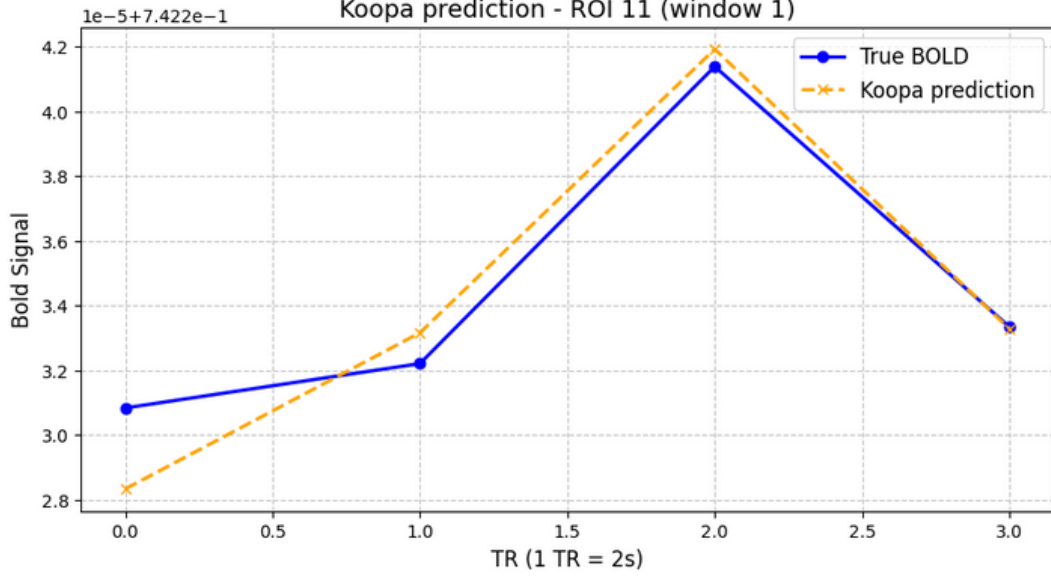


Figure 8: Visualisation Koopa prediction and true time series (ROI11)

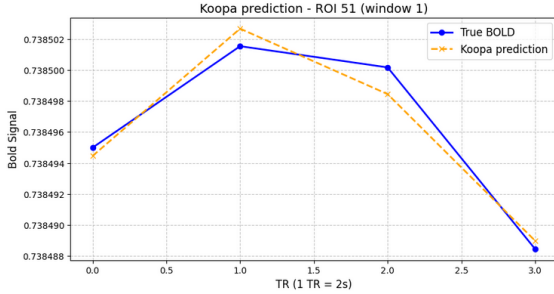


Figure 9: Visualisation Koopa prediction and true time series (ROI51)

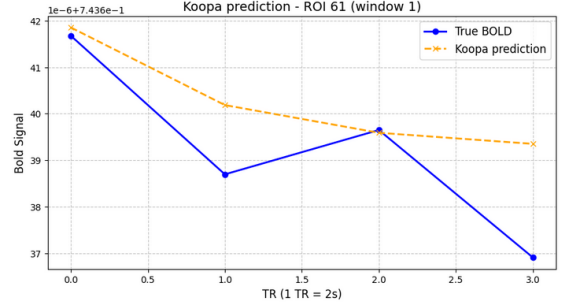


Figure 10: Visualisation Koopa prediction and true time series (ROI61)

Evaluation of the Koopa architecture reveals a robust ability to capture local dynamics and the physiological scale of the BOLD signal, while highlighting the inherent challenges in modeling high-frequency chaotic fluctuations. The quantitative results on the test set are summarized in Table 2.

Table 2: Global performance metrics on the test set

Metric	Value
MAE (Mean Absolute Error)	1.27×10^{-4}
MSE (Mean Squared Error)	3.79×10^{-8}
RMSE (Root Mean Squared Error)	1.95×10^{-4}
Mean Pearson Correlation (ρ)	0.512
Mean Directional Accuracy (MDA)	69.48%

Adherence to Physiological Scale (MAE and MSE) From a quantitative perspective, the model exhibits extremely low reconstruction errors (MAE of 1.27×10^{-4} and MSE of 3.79×10^{-8}). These results must be interpreted within the context of the simulated BOLD signals’ scale. Visual inspection of the predictions confirms that these scores do not result from a simple collapse of predictions toward the mean (variance collapse). On the contrary, the model rigorously adheres to the BOLD signal’s micro-scale, successfully adapting to amplitude variations on the order of 10^{-5} .

Capture of the Dynamic Envelope (MDA) The primary strength of the model lies in its modeling of the local temporal trend, objectified by a Mean Directional Accuracy (MDA) of 69.48%. The plots for ROIs 11 and 51 (see Figure 8) perfectly illustrate this metric: the predictive curve correctly anticipates rising and falling phases, faithfully following the inversion of peaks and troughs of the target signal. The model has thus effectively learned to deduce the sign of the local derivative.

Smoothing of Micro-fluctuations (Pearson Correlation) The moderate Pearson correlation (0.512) finds its visual explanation in the signal plots. For dynamics presenting abrupt slope discontinuities (as observed at the beginning of the window for ROI 61 10), the network manifests a slight inertia. While it correctly captures the overall downward trend, it tends to smooth out sudden drops. The model effectively restores the macroscopic and directional envelope of the signal but struggles to reproduce the exact complexity of high-frequency micro-fluctuations. This observation aligns with the HPO analysis (Section 3.2.4), which indicated the model’s insensitivity to fine spectral separation.

4.1.2 Analysis of the Koopman Decomposition

This section analyzes the internal decomposition performed by the Koopa architecture to interpret the nature of the representations learned by the Koopman operator on BOLD signals.

Variance Distribution between Branches The study of the relative contribution of the two model branches reveals a clear dominance of the dynamic component:

- **Time-variant branch:** Explains **83.65%** of the total variance.
- **Time-invariant branch:** Explains **13.33%** of the variance (the remaining $\approx 3\%$ representing covariance).

This specific distribution is primarily a consequence of the optimal hyperparameter α identified during HPO, which was set to a very low value (**0.15**). In the Koopa framework, α controls the spectral filtering threshold. Such a low value significantly limits the variance delegated to the invariant branch, forcing it to process only a marginal fraction of the signal while the time-variant branch absorbs the majority of the non-stationary fluctuations.

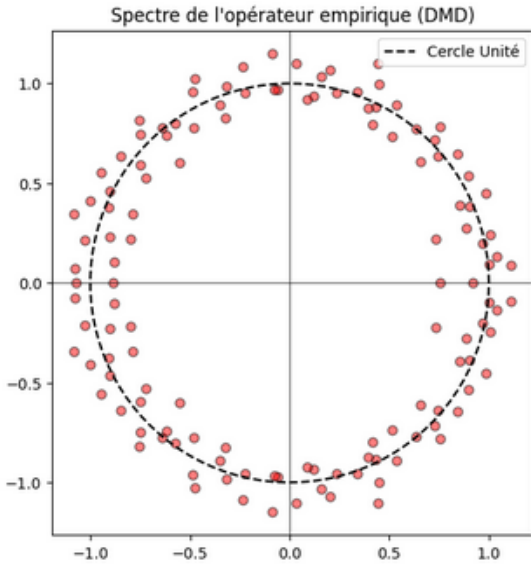


Figure 11: All Koopman mode in the complex plane

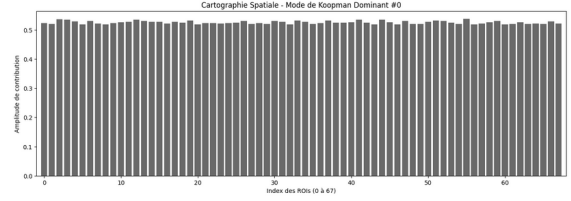


Figure 12: Contribution mode 0 to the 68 ROI

Spectral Analysis of the Invariant Operator The distribution of eigenvalues (λ) in the complex plane (Figure 11) provides a precise diagnostic of the learned dynamic regime:

- **Stable Oscillatory Regime:** Almost all eigenvalues are positioned in a ring around the unit circle ($|\lambda| \approx 1$). Mathematically, this indicates that the learned linear operator models a conservative system dominated by periodic or quasi-periodic cycles, without significant damping ($|\lambda| < 1$) or divergence ($|\lambda| > 1$).
- **BOLD Signature:** This spectral signature is typical of extracting fundamental rhythms of the BOLD signal, characterized by very low-frequency rhythmic fluctuations.
- **Approximation Noise:** The thickness of the ring and the points that are slightly offset from the circle confirm that the neural network (MLP) does not impose a strict structural constraint of energy conservation, resulting in a numerically imperfect approximation of the ideal Koopman operator.

Spatial Mapping and Dominant Mode (Mode #0) Projecting the latent state of the dominant eigenmode across the 68 regions of interest (ROIs) reveals a striking spatial uniformity:

- **Absence of Anatomical Specificity:** The contributions from all ROIs are almost identical. This Koopman mode fails to discriminate between localized functional networks (such as sensory or default mode networks), which invalidates the idea that the invariant operator captures complex spatial structures.
- **Global Signal Extraction:** In neuroimaging, such synchronous and uniform activity across all brain regions corresponds to the "**Global Signal.**" The invariant operator has essentially specialized in extracting the physiological baseline common to the entire brain (e.g., scanner drift, systemic vascular fluctuations).

4.1.3 Synthesis of Koopman Model Learning

Combining these analyses allows us to understand Koopa model and its strategy for processing TVB data:

1. **Task Delegation:** The spectral separation mechanism, constrained by the low α value, forces the invariant branch to act as a simple **global signal extractor**. It limits itself to modeling the rhythmic baseline observed in the spectrum and spatial mapping, which is coherent considering the heterogeneity of the data.
2. **Complexity Management:** Since the invariant Koopman operator fails to model transient local interactions or critical transitions, the network bypasses this theoretical limitation by relegating the information-bearing signal (the remaining variance) to the time-variant branch via local least-squares fitting.
3. **Impact of Channel Independence:** This spatial uniformity directly illustrates a structural limit of the architecture. By treating time series region-by-region in isolation, the invariant operator extracts a generic basis function for each ROI without successfully modeling the underlying functional connectivity matrix, a key factor in the dynamics of the simulated data.

5 Conclusion

The objective of this project was to model the complex spatiotemporal dynamics of brain activity as captured by functional Magnetic Resonance Imaging (fMRI). Specifically, this project sought to address the inherent challenges of fMRI time series, including inter-regional conduction delays and dynamic non-stationarity, by evaluating architectures capable of identifying the underlying governing equations of the signal.

Summary of Achievements The study relied on a methodological pipeline, beginning with the generation of biophysically grounded simulations using TheVirtualBrain (TVB). By employing an **AdEx mean-field model**, we produced simulated BOLD signals that reflect realistic metabolic and synaptic constraints. Following an study of these simulations, the **Koopman** architecture was implemented and optimized. The quantitative evaluation demonstrated a correct predictive performance, notably achieving a Mean Directional Accuracy (MDA) of nearly 70% and a correlation of over 0.51, proving the model’s ability to capture the macroscopic directional envelope of simulated BOLD fluctuations.

Critical Synthesis A deeper analysis of the model’s internal mechanics reveals a nuanced interaction between Koopman operator theory and deep learning components. The architecture successfully differentiates between invariant components, capturing background noise and signals shared across simulations, and a time-variant branch that effectively leverages Koopman theory through least-squares estimation.

However, the hyperparameter optimization (HPO) underscores a hierarchy of influence where the encoder-decoder (MLP) capacity remains the primary driver of predictive performance. The relative insensitivity of the final metric to the number of architectural blocks or the dimension of the lifted space suggests that the deep learning layers handle the bulk of the high-dimensional mapping. This implies that, for simulated BOLD time series, the Koopman framework serves as a coherent structural backbone, even if its theoretical advantages are partially overshadowed by the dominant role of the non-linear deep learning parameters.

Limitations and Future Perspectives Due to technical complexities and the time required for model calibration, several important areas remain to be done. The integration of **Graph Neural Networks (GNNs)** represents a next step to explicitly model the relational structure between regions and the structural delays of the functional connectivity. Furthermore, the

absence of testing on **empirical clinical data** (human or macaque) limits the immediate generalizability of these findings.

This project establishes a methodological framework and a comparative baseline. Future research should focus on models that incorporate an appropriate, non-stationary relational dimension between brain regions in the learning process in order to better predict brain dynamics.

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